§ 200. Normal units in H_m.

[source p. 728]

We now introduce a simplifying notation by putting

$$r^{5^\beta} = r_\beta, \tag{1}$$

and by letting β run through a complete system of residues modulo ν. It follows from this that

$$r_{\beta+\nu'} = -r_\beta, \tag{2}$$

and in the form (r, r_β), although not all substitutions of the group of the field Ω_m are then contained, since the substitutions (r, r_β^{-1}) must still be added, nevertheless (r, r_β) supplies us with all substitutions of the group of H_m.

We now denote by $\mathcal{E}(r)$ a unit of the field H_m. Among these are also contained the units of the field H_μ as imprimitive numbers, and they are characterized by the equation $\mathcal{E}(r) = \mathcal{E}(-r)$.

We shall now understand by a normal unit of the field H_m a unit which is not equal to ± 1, but which satisfies the equation

$$\mathcal{E}(r)\mathcal{E}(-r) = \pm 1. \tag{3}$$

It is clear that a unit of the field H_μ can in no case satisfy this condition. But not every unit in H_m which does not belong to the field H_μ will be a normal unit.[1]

The units τ_β [§ 199, (5)], however, belong to the normal units. For

$$\tau_\beta = \tan \frac{5^\beta \pi}{m} = \frac{1 - r_\beta}{1 + r_\beta} r_\beta^\nu, \tag{4}$$

and if in this the substitution $(r, -r) = (r_\beta, r_{\beta+\nu'})$ is made, then τ_β goes over into

$$\tau_{\beta+\nu'} = -\frac{1}{\tau_\beta}, \tag{5}$$

which corresponds to relation (3).

A system of ν' normal units

$$\mathcal{E}_0(r), \mathcal{E}_1(r), \ldots, \mathcal{E}_{\nu'-1}(r)$$

shall be called an independent system if the determinant

$$\pm \sum \pm \log|\mathcal{E}_0(r_0)| \log|\mathcal{E}_1(r_1)| \cdots \log|\mathcal{E}_{\nu'-1}(r_{\nu'-1})| = L(\mathcal{E}_0, \mathcal{E}_1, \ldots, \mathcal{E}_{\nu'-1}) \tag{6}$$

has a value different from zero.

We shall also here call the absolute value of this determinant, L, the regulator of the system $\mathcal{E}_0, \mathcal{E}_1, \ldots, \mathcal{E}_{\nu'-1}$.

The units $\tau_0, \tau_1, \ldots, \tau_{\nu'-1}$ form an independent system of normal units; for their regulator

$$L(\tau_0, \tau_1, \ldots, \tau_{\nu'-1})$$

[1] In the above-mentioned paper (Acta mathematica, vol. 8) the author called the normal units "primitive units". This name is rejected here because it could occasion the misunderstanding that every unit of the field H_m which does not at the same time belong to the field H_μ is to be counted among them.

is just the determinant T defined in the preceding paragraph [§ 199, (14), (15)]. And that this cannot be zero follows immediately from the fact that it occurs as a factor in the expression for the class number. But the class number, since at any rate at least one class, the principal class, exists, is different from zero, and so T also cannot be equal to zero. We have therefore the theorem:

1. The numbers $\tau_0, \tau_1, \dots, \tau_{\nu'-1}$ are an independent system of normal units of the field H_m.

If $\mathcal{E}(r)$ is a normal unit, then, if we put

$$L_\beta = \log |\mathcal{E}(r_\beta)|, \tag{7}$$

we have, because of (3),

$$L_\beta = -L_{\beta+\nu'}. \tag{8}$$

If now $\mathcal{E}_0, \mathcal{E}_1, \dots, \mathcal{E}_{\nu'-1}$ is any system of independent normal units, we put

$$L_{i,k} = \log |\mathcal{E}_i(r_k)|, \qquad L_{i,k+\nu'} = -L_{i,k}, \tag{9}$$

so that the regulator of the system of the $\mathcal{E}_i$ receives the expression

$$\pm \sum \pm L_{0,0} L_{1,1} \cdots L_{\nu'-1,\nu'-1} = L(\mathcal{E}_0, \mathcal{E}_1, \dots, \mathcal{E}_{\nu'-1}), \tag{10}$$

and is a number different from zero, which we denote by T_0.

If $\mathcal{E}(r)$ denotes any other normal unit, then the unknowns $\xi_0, \xi_1, \dots, \xi_{\nu'-1}$ can be determined from the ν' linear equations obtained when in

$$L_\beta = \xi_0 L_{0,\beta} + \xi_1 L_{1,\beta} + \cdots + \xi_{\nu'-1} L_{\nu'-1,\beta} \tag{11}$$

the index β is allowed to run from 0 to $\nu' - 1$; and, because of (8) and (9), this equation is also correct for the remaining values of β, so that from it one can derive the formula valid for all r,

$$\mathcal{E}(r) = \pm \mathcal{E}_0(r)^{\xi_0} \mathcal{E}_1(r)^{\xi_1} \cdots \mathcal{E}_{\nu'-1}(r)^{\xi_{\nu'-1}}. \tag{12}$$

On the other hand, if $m_0, m_1, \dots, m_{\nu'-1}$ denote arbitrary rational integers, then all numbers contained in the form

$$\pm \mathcal{E}_0(r)^{m_0} \mathcal{E}_1(r)^{m_1} \cdots \mathcal{E}_{\nu'-1}(r)^{m_{\nu'-1}}$$

are normal units of the field H_m. Accordingly, the considerations which we carried out in detail in § 186 can be applied to the system of equations (10), and from them it follows:

2. The exponents $\xi_0, \xi_1, \dots, \xi_{\nu'-1}$ determined from (10) or (11) are rational numbers.

[source p. 731]

From this one can infer, as in § 187, that there are systems of ν' independent normal units whose regulator T_0 is as small as possible, and such systems are called fundamental systems of normal units.

If $\mathcal{E}_0, \mathcal{E}_1, \dots, \mathcal{E}_{\nu'-1}$ is such a fundamental system, then the exponents $\xi_0, \xi_1, \dots, \xi_{\nu'-1}$ in (10) and (11) turn out to be rational integers. This too can be proved just as in § 187.

Under this supposition concerning the $\mathcal{E}_i$, if we set up equations (11) for any other system of independent units $\mathcal{E}_0^{(1)}, \mathcal{E}_1^{(1)}, \dots, \mathcal{E}_{\nu'-1}^{(1)}$, we obtain expressions of the form

$$L_{i,k}^{(1)} = \xi_{i,0} L_{0,k} + \xi_{i,1} L_{1,k} + \cdots + \xi_{i,\nu'-1} L_{\nu'-1,k}.$$

If then T_1 is the regulator of the system of the $\mathcal{E}_i^{(1)}$, then by the multiplication theorem for determinants, if C denotes the absolute value of the determinant of the integral exponents $\xi_{i,k}$, hence a positive rational integer, we get

$$T_1 = CT_0. \tag{13}$$

It cannot be proved, and for the higher values of m probably is not true, that $\tau_0, \tau_1, \dots, \tau_{\nu'-1}$ form a fundamental system.[2]

For our purpose, however, it is important that these units behave with respect to the denominator 2 in the exponents, in a certain sense, like a fundamental system; this is expressed more definitely by the following theorem:

2. If a normal unit of the field H_m of the form

$$\mathcal{E}(r) = \tau_0^{e_0} \tau_1^{e_1} \cdots \tau_{\nu'-1}^{e_{\nu'-1}},$$

in which the exponents $e_0, e_1, \dots, e_{\nu'-1}$ are rational integers, has the same sign in all its conjugate values, then all $e_0, e_1, \dots, e_{\nu'-1}$ must be even numbers, and $\mathcal{E}(r)$ is the square of a normal unit.

We can again conduct the proof by complete induction, but shall first transform the expression for τ_β somewhat.

The conjugate values of the unit $\mathcal{E}(r)$ we obtain in the form

$$\mathcal{E}(r_\beta) = \tau_\beta^{e_0} \tau_{\beta+1}^{e_1} \cdots \tau_{\beta+\nu'-1}^{e_{\nu'-1}},$$

and these expressions are therefore supposed to have the same sign for all values of β.

By § 199, (5),

$$\tau_\beta = \tan \frac{5^\beta \pi}{m} = \frac{1}{2} \frac{\sin\left(\frac{2\pi}{m} 5^\beta\right)}{\left[\cos\left(\frac{5^\beta \pi}{m}\right)\right]^2}.$$

Here, for abbreviation, we put

$$\sigma_\beta = \sin\left(\frac{2\pi}{m} 5^\beta\right), \tag{14}$$

so that σ_β always has the same sign as τ_β. From this we form the product

$$S_\beta = \sigma_\beta^{e_0} \sigma_{\beta+1}^{e_1} \cdots \sigma_{\beta+\nu'-1}^{e_{\nu'-1}}, \tag{15}$$

[2] Here, as in all questions concerning the units, it is very difficult to arrive at definite numerical results. According to the results contained in Reuschle's rich work, *Tafeln complexer Primzahlen, welche aus Wurzeln der Einheiten gebildet sind* (Berlin 1875), for $m = 16$ the τ still form a fundamental system. For here the ground number of the field H_m is equal to 2^{11}, the degree of the field is equal to 4, and therefore, by the note to § 156, in every ideal class there is an ideal whose norm is smaller than 6. According to Reuschle's tables, every natural prime number under 100 can however be decomposed in the field H_m into actually existing prime factors. Consequently in all numbers under 6 only principal ideals occur, and consequently there is here only the one class, the principal class. Since, as we have seen, the first class-number factor is also equal to 1, the cyclotomic field Ω_{16} is one-class. In it the same laws of prime numbers hold as in the field of rational numbers. This is doubtful for the field Ω_{32}, and certainly no longer true in the field Ω_{64}, in which the class number is a multiple of 17.

and our supposition is therefore that S_β has the same sign for all values of β.

It is to be proved that the integers $e_0, e_1, \dots, e_{\nu'-1}$ are all even.

The theorem is correct for $m = 8$; for in this case

$$\sigma_0 = \sin\frac{2\pi}{8}, \qquad \sigma_1 = \sin\frac{10\pi}{8} = -\sin\frac{6\pi}{8},$$

hence σ_0 is positive, σ_1 negative. Here $S_\beta = \sigma_\beta^{e_0}$, and therefore S_0 and S_1 can have the same sign only when e_0 is even.

We now apply complete induction. For simplification we put $\nu' = 2\nu''$ and recall the notation

$$m = 2^\varkappa, \qquad \mu = \frac{1}{2}m, \qquad \nu = \frac{1}{2}\mu, \qquad \nu' = \frac{1}{2}\nu, \qquad \nu'' = \frac{1}{2}\nu', \tag{16}$$

and suppose that our theorem has been proved as correct when μ takes the place of m.

Then

$$5^\nu \equiv 1 \pmod m, \quad 5^{\nu'} \equiv 1+\mu \pmod m, \quad 5^{\nu''} \equiv 1+\nu \pmod \mu,$$
$$5^{\nu''} \equiv 1+\nu+\mu \pmod m,$$

and from this it follows that

$$\sigma_{\beta+\nu'} = -\sigma_\beta, \qquad \sigma_{\beta+\nu''} = -\cos\left(\frac{2\pi}{m}5^\beta\right),$$

$$\sigma_\beta\sigma_{\beta+\nu''} = -\frac{1}{2}\sin\left(\frac{2\pi}{\mu}5^\beta\right) = -\frac{1}{2}\sigma'_\beta, \tag{17}$$

$$\sigma'_{\beta+\nu''} = -\sigma'_\beta, \tag{18}$$

where the σ'_β arise from the σ_β by the passage from $\varkappa$ to $\varkappa - 1$. Now, by (17), (18), we form the product $S_\beta S_{\beta+\nu''}$ and obtain, if for abbreviation we also put

$$e' = e_0 + e_1 + \cdots + e_{\nu''-1}, \qquad e'' = e_0 + e_1 + \cdots + e_{\nu'-1},$$

$$S_\beta S_{\beta+\nu''} = (-1)^{e''}\left(\frac{1}{2}\right)^{e'}(\sigma'_\beta)^{e_0+e_{\nu''}}(\sigma'_{\beta+1})^{e_1+e_{\nu''+1}}\cdots(\sigma'_{\beta+\nu''-1})^{e_{\nu''-1}+e_{\nu'-1}}.$$

This, however, is a product of the same form as S_β, in which μ has taken the place of m, and it also has the same sign with all its conjugates. Therefore, by the assumption made,

$$e_0 \equiv e_{\nu''}, \quad e_1 \equiv e_{\nu''+1}, \quad \dots, \quad e_{\nu''-1} \equiv e_{\nu'-1} \pmod 2. \tag{19}$$

Thus putting

$$S'_\beta = (\sigma'_\beta)^{e_0}(\sigma'_{\beta+1})^{e_1}\cdots(\sigma'_{\beta+\nu''-1})^{e_{\nu''-1}},$$
$$R_\beta = \sigma_{\beta+\nu''}^{\frac{1}{2}(e_0-e_{\nu''})}\sigma_{\beta+\nu''+1}^{\frac{1}{2}(e_1-e_{\nu''+1})}\cdots\sigma_{\beta+\nu'-1}^{\frac{1}{2}(e_{\nu''-1}-e_{\nu'-1})},$$

we get from (15) and (17)

$$S_\beta R_\beta^2 = \left(-\frac{1}{2}\right)^{e'} S'_\beta,$$

and hence it follows that S'_β, just like S_β, has the same sign with all its conjugate values. But S'_β arises from S_β by putting μ in place of m, and by our assumption therefore

$$e_0 \equiv 0, \quad e_1 \equiv 0, \quad \dots, \quad e_{\nu''-1} \equiv 0 \pmod 2,$$

and with the help of (19)

$$e_{\nu''} \equiv 0, \quad e_{\nu''+1} \equiv 0, \quad \ldots, \quad e_{\nu'-1} \equiv 0 \pmod 2,$$

which proves theorem 2.

If now in formula (12) for $\mathcal{E}_0, \mathcal{E}_1, \ldots, \mathcal{E}_{\nu'-1}$ we choose the system $\tau_0, \tau_1, \ldots, \tau_{\nu'-1}$ and put the exponents ξ in the form

$$\xi_0 = \frac{e_0}{e}, \quad \xi_1 = \frac{e_1}{e}, \quad \ldots, \quad \xi_{\nu'-1} = \frac{e_{\nu'-1}}{e},$$

where $e, e_0, e_1, \ldots, e_{\nu'-1}$ are rational integers without a common divisor, then we obtain

$$\mathcal{E}(r)^e = \pm \tau_0^{e_0} \tau_1^{e_1} \cdots \tau_{\nu'-1}^{e_{\nu'-1}},$$

and from our theorem it follows that e must be odd; for if it were even, $\mathcal{E}(r)^e$ would be the square of a real quantity, and consequently $\pm\mathcal{E}(r)^e$ would have the same sign with its conjugates. But then, by our theorem, all $e_0, e_1, \ldots, e_{\nu'-1}$ would have to be divisible by 2, which again contradicts the supposition that the exponents $e, e_0, \ldots, e_{\nu'-1}$ should have no common divisor. We have therefore the theorem:

3. If $\mathcal{E}(r)$ is any normal unit of the field H_m, then the odd number e and the integers $e_0, e_1, \ldots, e_{\nu'-1}$ can be so determined that

$$\mathcal{E}(r_\beta)^e = \pm \tau_\beta^{e_0} \tau_{\beta+1}^{e_1} \cdots \tau_{\beta+\nu'-1}^{e_{\nu'-1}}$$

holds.

This result leads to some important consequences:

4. If T is the regulator of the system of the τ_β, and T_0 the regulator of a fundamental system of normal units in H_m, i.e. the minimum value that the regulator of any system of independent normal units can assume, then

$$T = CT_0, \qquad C \equiv 1 \pmod 2, \tag{21}$$

where C denotes an odd natural number.

[source p. 735]

Indeed, that $T : T_0 = C$ is an integer follows from formula (13).

But if we apply theorem 3 to all elements of the fundamental system $\mathcal{E}_i(r)$, there results a system of integers $e_{i,k}$ and an odd number e, such that

$$\mathcal{E}_i(r)^e = \pm \tau_0^{e_{i,0}} \tau_1^{e_{i,1}} \cdots \tau_{\nu'-1}^{e_{i,\nu'-1}}.$$

Taking logarithms of this and then forming the regulator T_0, we obtain

$$e^{\nu'} T_0 = \pm \sum \pm e_{0,0} e_{1,1} \cdots e_{\nu'-1,\nu'-1} T,$$

hence

$$e^{\nu'} = \pm C \sum \pm e_{0,0} e_{1,1} \cdots e_{\nu'-1,\nu'-1},$$

and since the left side here is odd, C cannot be even, as was to be proved.

If all conjugate values of the normal unit $\mathcal{E}(r_\beta)$ have the same sign, then the same is true of $\mathcal{E}(r_\beta)^e$, and consequently in this case in formula (20) the exponents $e_0, e_1, \ldots, e_{\nu'-1}$ must be even according to 2. If one writes formula (20) in the form

$$\mathcal{E}(r) = \pm \tau_0^{e_0} \tau_1^{e_1} \cdots \tau_{\nu'-1}^{e_{\nu'-1}} \mathcal{E}(r)^{1-e},$$

one obtains from it the theorem:

5. A normal unit of the field H_m which has the same sign with its conjugates is, apart from the sign, a square of a normal unit.

§ 201. Fundamental system of units of the field H_m.

There remains still to determine the denominator D in the formula for B [§ 197, (6)], which was defined by the equation

$$E = E'D,$$

where E and E' denoted the regulators of the fields H_m, H_μ.

Here, therefore, the issue is no longer the normal units, but the units of the field H_m in general.

[source p. 736]

We assume a fundamental system of units in the field H_m:

$$\varepsilon_1(r), \varepsilon_2(r), \ldots, \varepsilon_{\nu-1}(r), \tag{1}$$

and let the conjugate logarithms of this system be

$$l_{1,k}, l_{2,k}, \ldots, l_{\nu-1,k}, \qquad k = 0, 1, 2, \ldots, \nu - 1. \tag{2}$$

Beside this we consider a fundamental system in the field H_μ,

$$\varepsilon'_1, \varepsilon'_2, \ldots, \varepsilon'_{\nu'-1}, \tag{3}$$

with the conjugate logarithms

$$l'_{1,k}, l'_{2,k}, \ldots, l'_{\nu'-1,k}, \qquad k = 0, 1, 2, \ldots, \nu' - 1. \tag{4}$$

In forming the regulators one of the conjugate values, say $k = 0$, is omitted (compare § 186), and therefore we obtain

$$\begin{aligned} E &= \pm \sum \pm l_{1,1} l_{2,2} \cdots l_{\nu-1,\nu-1}, \\ E' &= \pm \sum \pm l'_{1,1} l'_{2,2} \cdots l'_{\nu'-1,\nu'-1}. \end{aligned} \tag{5}$$

Now to the system (3) we add a fundamental system of normal units of the field H_m,

$$\mathcal{E}_0, \mathcal{E}_1, \ldots, \mathcal{E}_{\nu'-1}, \tag{6}$$

and we assert that the system

$$\varepsilon'_1, \varepsilon'_2, \ldots, \varepsilon'_{\nu'-1}, \mathcal{E}_0, \mathcal{E}_1, \ldots, \mathcal{E}_{\nu'-1} \tag{7}$$

is an independent system of units of the field H_m (although not a fundamental system).

If we denote by

$$L_{0,k}, L_{1,k}, \ldots, L_{\nu'-1,k}, \qquad k = 0, 1, \ldots, \nu' - 1, \tag{8}$$

the conjugate logarithms of the system (6), then by the preceding paragraph

$$\pm \sum \pm L_{0,0} L_{1,1} \cdots L_{\nu'-1,\nu'-1} = T_0 \tag{9}$$

is the regulator of this system, and we have, with regard to the definition of the normal units [§ 200, (9)],

$$L_{i,k+\nu'} = -L_{i,k}, \tag{10}$$

and because the ε'_i as numbers of the field H_μ remain unaltered by the substitution $(r, -r)$,

$$l'_{i,k+\nu'} = l'_{i,k}. \tag{11}$$

[source p. 737]

Accordingly the regulator R of the system (7) is obtained from the determinant

$$\begin{vmatrix} l'_{1,1} & l'_{2,1} & \cdots & l'_{\nu'-1,1} & L_{0,1} & L_{1,1} & \cdots L_{\nu'-1,1} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ l'_{1,\nu'-1} & l'_{2,\nu'-1} & \cdots & l'_{\nu'-1,\nu'-1} & L_{0,\nu'-1} & L_{1,\nu'-1} & \cdots L_{\nu'-1,\nu'-1} \\ l'_{1,0} & l'_{2,0} & \cdots & l'_{\nu'-1,0} & -L_{0,0} & -L_{1,0} & \cdots - L_{\nu'-1,0} \\ l'_{1,1} & l'_{2,1} & \cdots & l'_{\nu'-1,1} & -L_{0,1} & -L_{1,1} & \cdots - L_{\nu'-1,1} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ l'_{1,\nu'-1} & l'_{2,\nu'-1} & \cdots & l'_{\nu'-1,\nu'-1} & -L_{0,\nu'-1} & -L_{1,\nu'-1} & \cdots - L_{\nu'-1,\nu'-1} \end{vmatrix}.$$

This determinant has $\nu - 1$ rows; we add the first to the $(\nu' + 1)$-st, the second to the $(\nu' + 2)$-nd, and so on, the $(\nu' - 1)$-st to the last. Thereby the L disappear in the last $(\nu' - 1)$ rows, and the l receive the factor 2 [by (10), (11)], and from this, by a theorem on determinants (Vol. I, § 23, V), it follows that

$$R = 2^{\nu'-1} E' T_0. \tag{12}$$

Since according to this R does not vanish, and therefore in (7) there is a system of $\nu - 1$ independent units of the field H_m, we can [by § 186, (5)] determine the rational (integral or fractional) numbers $m_{i,k}, M_{i,k}$ from the equations

$$2l_{i,k} = m_{1,i} l'_{1,k} + \cdots + m_{\nu'-1,i} l'_{\nu'-1,k} + M_{0,i} L_{0,k} + M_{1,i} L_{1,k} + \cdots + M_{\nu'-1,i} L_{\nu'-1,k}. \tag{13}$$

It is, however, easy to prove from the relations (10), (11) that the numbers $m_{i,k}, M_{i,k}$ here must be integers. For

$$l_{i,k} + l_{i,k+\nu'} = \log|\varepsilon_i(r_k)\varepsilon_i(-r_k)| = m_{1,i} l'_{1,k} + \cdots + m_{\nu'-1,i} l'_{\nu'-1,k},$$

$$l_{i,k} - l_{i,k+\nu'} = \log|\varepsilon_i(r_k)\varepsilon_i(-r_k)^{-1}| = M_{0,i} L_{0,k} + M_{1,i} L_{1,k} + \cdots + M_{\nu'-1,i} L_{\nu'-1,k}.$$

The units $\varepsilon_i(r)\varepsilon_i(-r)$ belong however to the field H_μ, and since the ε'_i were by supposition a fundamental system of this field, the $m_{i,k}$ must be rational integers by § 187.

The units $\varepsilon_i(r)\varepsilon_i(-r)^{-1}$ are normal units of the field H_m, and since the $\mathcal{E}_i$ were supposed to be a fundamental system of normal units, the $M_{i,k}$ are integers by § 200.

[source p. 738]

For better survey we set down the equations (13) once more without the index k, which distinguishes the conjugate values from one another:

$$2l_i = m_{1,i} l'_1 + m_{2,i} l'_2 + \cdots + m_{\nu'-1,i} l'_{\nu'-1} + M_{0,i} L_0 + M_{1,i} L_1 + M_{2,i} L_2 + \cdots + M_{\nu'-1,i} L_{\nu'-1}. \tag{14}$$

If now, according to (13), we form the regulator E, and denote by M the absolute value of the determinant of the $\nu - 1$ rows of the numbers $m_{k,i}, M_{k,i}$, then it follows that

$$2^{\nu-1} E = MR, \tag{15}$$

and from this, with regard to (12),

$$E = 2^{-\nu'} M E' T_0. \tag{16}$$

It can now further be shown that M is a power of 2 and is divisible by $2^{\nu'}$.

For since the system (1) was supposed to be a fundamental system in H_m, there are rational integers $n_{k,i}, N_{k,i}$ satisfying the equations

$$\begin{aligned} l'_i &= n_{1,i}l_1 + n_{2,i}l_2 + \cdots + n_{\nu-1,i}l_{\nu-1}, & i &= 1, 2, \ldots, \nu' - 1, \\ L_i &= N_{1,i}l_1 + N_{2,i}l_2 + \cdots + N_{\nu-1,i}l_{\nu-1}, & i &= 0, 1, 2, \ldots, \nu' - 1. \end{aligned} \tag{17}$$

If we form from this the determinant R and denote by N the absolute value of the determinant of the $n_{k,i}, N_{k,i}$, we get

$$R = NE,$$

and from this by (15),

$$MN = 2^{\nu-1};$$

therefore each of the two integers M, N must be a power of 2.

In order to decide by what power of 2 the determinant M is divisible, we first determine a system of rational integers a_i without a common divisor which satisfies the linear equations

$$\sum_{i=1}^{\nu-1} a_i m_{k,i} = 0 \qquad (k = 1, 2, \ldots, \nu' - 1). \tag{18}$$

If we then put

$$\sum_{i=1}^{\nu-1} a_i M_{k,i} = \xi_k,$$

[source p. 739]

then from equations (13) we obtain

$$2\sum_{i=1}^{\nu-1} a_i l_{i,k} = \xi_0 L_{0,k} + \xi_1 L_{1,k} + \cdots + \xi_{\nu'-1} L_{\nu'-1,k}.$$

It follows from this, by (10), that

$$\sum_{i=1}^{\nu-1} a_i l_{i,k+\nu'} = -\sum_{i=1}^{\nu-1} a_i l_{i,k},$$

and this means that

$$\delta = \varepsilon_1^{a_1}\varepsilon_2^{a_2}\cdots\varepsilon_{\nu-1}^{a_{\nu-1}}$$

is a normal unit of the field H_m. From this it then follows further, because $\mathcal{E}_0, \mathcal{E}_1, \ldots, \mathcal{E}_{\nu'-1}$ is a fundamental system of normal units, that the numbers $\xi_0, \xi_1, \ldots, \xi_{\nu'-1}$ must be even. Hence:

1. The system of equations (18) has the congruences

$$\sum_{i=1}^{\nu-1} a_i M_{k,i} \equiv 0 \pmod 2, \qquad k = 0, 1, \ldots, \nu' - 1, \tag{19}$$

as consequences.

In equations (18), however, whose number is only $\nu' - 1$, there are $\nu - 1$ unknowns a_i. Therefore there are several systems of independent solutions, which we can state a little more precisely as follows:

2. There are ν' integral solutions of the system (18),

$$\begin{array}{cccc} a_{1,1}, & a_{2,1}, & \ldots, & a_{\nu-1,1}, \\ a_{1,2}, & a_{2,2}, & \ldots, & a_{\nu-1,2}, \\ \cdots & \cdots & \cdots & \cdots \\ a_{1,\nu'}, & a_{2,\nu'}, & \ldots, & a_{\nu-1,\nu'}, \end{array} \tag{20}$$

of such a kind that from the matrix (20) a determinant of ν' rows can be formed which is an odd number.

For first there is a solution of (18)

$$a_{1,1}, a_{2,1}, a_{3,1}, \ldots, a_{\nu-1,1},$$

in which an odd number occurs. Suppose, as is allowed, since here nothing depends on the ordering of the indices, that $a_{1,1}$ is this odd number; then we determine a second solution

$$0, a_{2,2}, a_{3,2}, \ldots, a_{\nu-1,2},$$

in which again an odd number, say $a_{2,2}$, occurs. Then we determine a third solution

$$0, 0, a_{3,3}, \ldots, a_{\nu-1,3},$$

and continue setting zero so long as the number of unknowns remaining is still greater than the number of equations.

In the last system, therefore, $\nu' - 1$ zeros will occur, so that ν' unknowns remain.

We can put these rows of numbers $a_{\nu,k}$ for the system (20), for in it the determinant

$$\pm \sum \pm a_{1,1} a_{2,2} \cdots a_{\nu',\nu'} \tag{21}$$

reduces to the diagonal term $a_{1,1} a_{2,2} \cdots a_{\nu',\nu'}$, whose factors are all odd.

If now the requirement of theorem 2 is fulfilled, the matrix (20) can be completed by adjoining $\nu' - 1$ rows to a determinant of $\nu - 1$ rows which likewise has an odd integral value:

$$a = \pm \sum \pm a_{1,1} a_{2,2} \cdots a_{\nu',\nu'} a_{\nu'+1,\nu'+1} \cdots a_{\nu-1,\nu-1}. \tag{22}$$

One need only, when the determinant (21) is assumed odd, put zeros throughout in the added rows, except for the elements standing in the diagonal row, $a_{\nu'+1,\nu'+1} \cdots a_{\nu-1,\nu-1}$, which may be taken equal to 1. Then a receives the same value as the determinant (21).

But if now, by the multiplication theorem for determinants, we form the product aM, a determinant results in which the $\nu - 1$ sums

$$\sum_{i=1}^{\nu-1} a_{i,s} m_{k,i}, \qquad \sum_{i=1}^{\nu-1} a_{i,s} M_{k,i}$$

$$k = 1, 2, \ldots, \nu' - 1, \qquad k = 0, 1, 2, \ldots, \nu' - 1,$$

form a row for every value $s = 1, 2, \ldots, \nu'$, and in which, therefore, ν' rows occur whose elements are all divisible by 2.

[source p. 741]

Accordingly aM, and consequently also M, is divisible by $2^{\nu'}$, and if we put

$$M = 2^{\nu'+\sigma}, \qquad N = 2^{\nu-\sigma-1}, \tag{23}$$

then σ is a non-negative rational integer.

If we insert this in (16), it follows that

$$E = 2^\sigma E' T_0.$$

But in § 197, (6),

$$E = E' D$$

was put, and therefore there results

$$D = 2^\sigma T_0.$$

From this there follows for the class-number factor B, by § 199, (15) and § 200, (21),

$$B = 2^{-\sigma} \frac{T}{T_0} = 2^{-\sigma} C, \tag{24}$$

where C is an odd integer.

Accordingly, by § 197, (4), for the second factor of the class number h_1, which, as we have seen, is an integer, there results

$$h_1 = 2^{-\sigma} h_1' C. \tag{25}$$

If we now assume as proved that h_1' is odd, then, since C is also odd, it follows that $\sigma = 0$ and h_1 is odd; and with this both are proved by complete induction.

Thus the results derived are simplified as follows:

3. One has

$$M = 2^{\nu'}, \qquad D = T_0, \qquad B = \frac{T}{T_0}; \tag{26}$$

the class-number factor B is equal to the ratio of the regulator T of the units τ to the regulator T_0 of a fundamental system of normal units, and is an odd integer.

If we denote by $B_\varkappa$ the number B belonging to $m = 2^\varkappa$, then the class number h_1, as follows by complete induction, is

$$h_1 = B_4 B_5 \cdots B_\varkappa, \tag{27}$$

and is therefore also an odd number. With this, with regard to § 198, 1, the principal theorem is proved:

A. The class number in the field Ω_m, when m is a power of 2, is odd.

[source p. 742]

§ 202. Positive units.

The results last found show that the system of independent units § 201, (7),

$$\varepsilon_1', \varepsilon_2', \ldots, \varepsilon_{\nu'-1}', \mathcal{E}_0, \mathcal{E}_1, \ldots, \mathcal{E}_{\nu'-1}$$

does not form a fundamental system in the field H_m. For otherwise in the formulae § 201, (13) the numbers $m_{k,i}, M_{k,i}$ would all have to be divisible by 2, and the determinant M, whose value we found to be $2^{\nu'}$, would be divisible by $2^{\nu-1}$.

We now consider, instead of the system of equations (18), § 201, the following congruences:

$$\begin{aligned}
\sum_{i=1}^{\nu-1} b_i m_{1,i} &\equiv 1 \pmod 2, \\
\sum_{i=1}^{\nu-1} b_i m_{2,i} &\equiv 0 \pmod 2, \\
&\vdots \\
\sum_{i=1}^{\nu-1} b_i m_{\nu'-1,i} &\equiv 0 \pmod 2,
\end{aligned} \tag{1}$$

and prove that it is possible to satisfy them by integral values of the b_i.

If this were not possible, then from the $\nu' - 2$ congruences

$$\sum_{i=1}^{\nu-1} b_i m_{k,i} \equiv 0 \pmod 2 \qquad k = 2, 3, \dots, \nu' - 1$$

the last one

$$\sum_{i=1}^{\nu-1} b_i m_{1,i} \equiv 0 \pmod 2$$

would have to result as a consequence; and then, as in the preceding paragraph, it would further follow that

$$\sum_{i=1}^{\nu-1} b_i M_{k,i} \equiv 0 \pmod 2, \qquad k = 0, 1, \dots, \nu' - 1.$$

[source p. 743]

Then one could put in place of the matrix § 201, (20) a matrix of the $b_{k,i}$ of $\nu' + 1$ rows, and it could be concluded on the path taken in the preceding paragraph that M would have to be divisible by $2^{\nu'+1}$, which is not the case.

If however we apply the system of multipliers b_i, which satisfies conditions (1), to equations (14) of the preceding paragraph, by multiplying by b_i and then summing, then there results, if we leave out multiples of 2 and express this also here, where irrational numbers, namely logarithms, occur, by the sign of congruence,

$$l'_1 \equiv L_0 \sum M_{0,i} b_i + L_1 \sum M_{1,i} b_i + \cdots + L_{\nu'-1} \sum M_{\nu'-1,i} b_i \pmod 2. \tag{2}$$

If we want to transform the congruence into an equation, a sum of quantities

$$l_1, l_2, \dots, l_{\nu-1}, \qquad l'_1, l'_2, \dots, l'_{\nu'-1}$$

with even integers as coefficients is to be added.

The same consideration can however be applied to $l'_2, l'_3, \dots, l'_{\nu'-1}$; and then, when one passes from the logarithms back to the numbers, one may express formula (2) in words as follows:

1. Every unit of the field H_μ is representable as the product of a normal unit of the field H_m and a square of a unit in H_m.

If we now apply theorem 2 of § 200 to this, it results that every unit $\mathcal{E}(r)$ of the field H_μ which has the same sign with all its conjugates must, apart from its sign, be the square of a unit in H_m. If however

$$\pm\mathcal{E}(r) = \Delta(r)^2,$$

where $\mathcal{E}(r)$ is a unit in H_μ and $\Delta(r)$ is a unit in H_m, then $\Delta(r)$ must also be contained in H_μ.

For put

$$\Delta(r) = \Delta_1(r^2) + r\Delta_2(r^2).$$

Then the square of this can be contained in H_μ, and so remain unchanged by the change of sign of r, only if either Δ_1 or Δ_2 is zero. But $\Delta(r) = r\Delta_2(r^2)$ cannot hold; for then $\Delta_2(r^2)$ would be a unit in Ω_μ and, by § 178, 1, would have to be the product of a real unit and a root of unity $r^{2\lambda}$; and since $\Delta(r)$ is also real, $r^{2\lambda+1}$ would have to be real, which is impossible. Therefore $\Delta(r)$ is contained in H_μ.

Since now μ, just like m, can be any power of 2, there follows from this the second principal theorem:

B. A unit of the field H_m which is positive with all its conjugates is the square of a unit of the same field.

On the two theorems A and B, however, in § 184 we made the proof of the principal theorem on the Abelian number fields (§ 179, I) depend.